FTTH/GPON Fault Diagnostics and Field Operations Improvement

Portfolio Case Study
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Focus Areas: Fibre Diagnostics, GPON, ONT/CPE, Incident Management, SLA Tracking

1. Project Overview

This project demonstrates a structured approach to diagnosing fibre access network faults, restoring customer connectivity, and documenting technical evidence for FTTH/GPON service providers. It combines practical fibre troubleshooting workflows with incident management, escalation procedures, and service-level reporting to improve field-service quality and first-visit resolution rates.

2. Network Topology & Fault Domains

(Reference Diagram: FTTH/GPON Network Architecture)
The FTTH/GPON network is divided into two primary segments: the Outside Plant (OSP) and Customer Premises.

- **Outside Plant (OSP):** The OLT, Feeder Fibre, Optical Splitter, Distribution Point (FDH), and Drop Cable. The most common faults in this domain are signal loss from the source, high loss/wrong split ratios, dirty connectors, and physical cable damage.

- **Customer Premises:** The ONT, Customer Router, and Wi-Fi Devices. Common faults include Loss of Signal (LOS) alarms, provisioning failures, DHCP misconfigurations, and Wi-Fi interference.

3. Troubleshooting Methodology

(Reference Diagram: Fault-Isolation Flowchart)
To avoid unnecessary equipment replacement and repeat site visits, a standardized diagnostic workflow is essential.

- **Scenario A (Physical Layer):** If the ONT shows a Red LOS light, the technician must check the optical power meter. A reading below -27 dBm indicates an issue with the drop cable, connector contamination, or an upstream splitter fault. If a clean connection still yields low power, the incident is escalated to the OSP team with documented evidence.
- **Scenario B (Provisioning):** If the ONT has power but the PON light is blinking or off, the issue is likely ONT registration. The technician verifies the serial number and provisioning status in the OLT.
- **Scenario C (Wireless):** If Ethernet connectivity is stable but Wi-Fi is slow, the fault is localized to the router. The technician checks channel utilization, 2.4GHz vs 5GHz band selection, and physical placement to eliminate interference.

4. Incident Management & KPI Tracking

(Reference Document: Excel KPI Dashboard)
Field operations require strict adherence to Service Level Agreements (SLAs). I developed an Incident Log and KPI Dashboard (simulated data) to track operational performance.

Key Performance Indicators (KPIs) tracked:

- **Average Time to Restore:** Identifies technician efficiency and bottleneck areas.
- **SLA Compliance Rate:** Tracks the percentage of tickets resolved within the mandated 6-hour window.
- **Fault Domain Distribution:** Highlights recurring infrastructure issues (e.g., multiple drop-cable faults in one neighborhood), allowing for proactive OSP maintenance rather than reactive repairs.

5. Conclusion

By implementing a structured fault-isolation decision tree, consistent optical measurement logs, and strict SLA tracking, telecom operations can reduce repeat complaints, improve first-visit resolution rates, and maintain high customer satisfaction (CSAT). This case study reflects my approach to delivering Tier-1/Tier-2 field operations and incident management.